

Vladimir Bronnikov (Dr. Bronn)

PhD. Computer Science&Robotics

An initial interest in AI arose while writing a dissertation on visual pattern recognition for robots. A text-reading system was developed using a phototelegraphic device and a PDP-8 computer.

From 1980, he studied biophysics and neuroscience, lectured, and led the development of neurocomputers within the Mathematical Biology group at St. Petersburg University (USSR).

Starting in 1981, he developed the theory of generalized cellular automata (CA) and its application to modeling biological systems, including various types of neural networks.

In 1986, as a result of research into CA models, he created a new abstract game called CTOR (Cellular Torus); it gained popularity in Russia starting in the late 1990s and saw a production run of 100,000 copies.

From 1990, he led the development of the "Locon" supercomputer—based on 16,000 processors with a parallel architecture—at a design bureau (LKTb, Leningrad, USSR, with the support of I.V. Berg). Details of this development were published in the article "Avionics at the Turn of the Millennium" in the journal *Mir Avioniki* (Issues 1–2, 2000, pp. 45–49).

He is the author of scientific and popular science articles, as well as the monograph *Mathematical Modeling of Biosystems* (St. Petersburg State University, former LGU 1989).

I currently reside in Canada, where I continue to develop the mechanics of the CTOR game and explore their application across various fields—from sports to science and education. In 2016, I gave a TEDx talk titled "How to Measure Collective Thinking?"

This area of research—specifically the collective behavior of agents—is what interests me most today, and it represents one of the key challenges in developing AI-agent-based systems. We are confident that our game mechanics model will help solve this problem. We are open to collaboration, so please feel free to reach out.

A list of selected publications and their connection to contemporary AI issues can be found here:

Vladimir Bronnikov, PhD | Academic & Research Portfolio

**Foundational Research in Pattern Recognition,
Cellular Automata, Supercomputer Architectures, and Collective Intelligence.**

Section 1: Educational Experience/"Computational Biology."

1. University Textbook: "Introduction to Mathematical Methods and Programming for Biologists"

- Original Citation: Bronnikov, V. A. Introduction to Mathematical Methods and Programming for Biologists: A Textbook / A. A. Zhdanov Leningrad State University. — Leningrad: Leningrad State University, 1987. — 83 p.

- **Abstract:** This textbook presents mathematical modeling methods for the emerging field of computational biology. It replaces the mathematics of differential equations with computable cellular automaton models implemented as program code. This approach serves as a foundation for studying biological systems at various levels—ranging from cell populations, predator-prey models, and ecosystems to neural networks. The book provides examples of how biological processes are formalized and offers corresponding software solutions.
- **Bridge to CTOR:** This work represents the first attempt to translate the laws of biological interaction into program code. The developed methods have paved the way for modeling biological systems with complex topological structures and emergent system behavior.

Section 2: Foundational Research in Pattern Recognition & Computer Science

2. Doctoral Dissertation: "Pattern Recognition of Drawing and Graphical Images for Computer Input"

- **Original Citation:** Bronnikov, V. A. Recognition of Technical Drawings and Graphical Images During Computer Input. Dissertation for the degree of Candidate of Technical Sciences. V. I. Ulyanov (Lenin) Leningrad Electrotechnical Institute (LETI), Department of Computer Engineering, 1980.
- **Abstract:** This doctoral dissertation addresses a specific problem in computer vision: formalizing the detection of connectivity within planar graphical objects, such as technical drawings and maps. A novel method was employed—specifically, pushdown automata (an extension of the Turing machine model). The study details the development of algorithms capable of processing and decoding structural graphical data received from facsimile devices, and a functional prototype of the reading machine was successfully implemented. The total volume of code written amounted to 20,000 instructions in PDP-8 Assembly language.
- **Bridge to CTOR:** The core algorithmic constraints developed for identifying continuous graphical paths and boundary tracking directly influenced the mathematical formalization of spatial perception and pattern-matching rules embedded within the matrix logic of CTOR.

3. Conference Paper: "Application of Fuzzy Sets in Pattern Recognition"

- **Original Citation:** Bronnikov, V. A. Use of Fuzzy Sets for Image Recognition. Proceedings of the All-Union Conference, Kaunas, 1979.
- **Abstract:** This paper presents a method of applying Lotfi Zadeh's Fuzzy Set Theory to algorithmic image processing. By modeling visual thresholds as membership functions rather than binary logic gates, the proposed system achieved significantly higher error tolerance in noisy robotic environments.
- **Bridge to CTOR:** The transition from binary states to probabilistic/fuzzy interactions underlies the advanced game mechanics of CTOR, where player interactions simulate dynamic, adaptive shifting rather than static combinatorial outcomes.

Section 3: Peer-Reviewed Scientific Papers (Cellular Automata & Neural Computing)

4. Journal Paper: "Identification of Additive Cellular Automata"

- Original Citation: Adamatzky, A. I., Bronnikov, V. A. **Identification of additive cellular automata** // Izvestiya AN SSSR. Series "Technical Cybernetics", 1989, Vol. 3, pp. 200–205. (English parallel publication tracked as Technical Cybernetics, 1989).
- Abstract: Co-authored with Andrew Adamatzky, this paper explores the reverse-engineering problem of cellular automata (CA). It presents an analytical framework to identify the underlying local transition rules of additive CA based strictly on their macroscopic temporal behaviors and evolution history.
- Bridge to CTOR: This is the mathematical birthplace of CTOR. The realization that highly complex global behaviors could be mathematically driven (and identified) by simple, bounded local transition rules led directly to using a boundaryless 2D field (Cellular Torus) to test strategic agent behaviors.

5. International Journal Publication: "Graphic Presentation of Nucleotide Sequences in Computer Simulation on Gene Evolution"

- Original Citation: Adamatzky, A. I., & Bronnikov, V. A. **Graphic presentation of nucleotide sequences in computer simulation on gene evolution** // *Studia Biophysica*, 1989, v. 125, p. 251-255.
- Abstract: This peer-reviewed paper outlines a methodology for mapping genetic code (DNA/RNA nucleotide sequences) onto spatial-graphic environments. By simulating evolutionary algorithms within a computer network, the research explores patterns of gene selection and mutation propagation through geometric modeling.
- Bridge to CTOR: This study confirms CTOR's lineage as a biological modeling tool. The representation of genetic data as dynamic visual structures on a grid mirror how CTOR tokens function as living entities competing within a closed ecosystem.
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6. Scientific Book Chapter: "Neural Nets Simulation by Cellular Automata with Global Feedback"

- Original Citation: Bronnikov, V. A., & Adamatzky, A. I. **Neural nets simulation by cellular automata with global feedback** // In: *NEURONET '90*, Prague, 1990, p. 58-59.
- Abstract: This is a pioneering work that introduced the concept of "global connectivity" into the theory of cellular automata. It opened up new possibilities for this—at the time—underappreciated mathematical framework, which was subsequently further developed by Andrew Adamatzky (a student of the author) and Stephen Wolfram (a colleague from the USA).
- Bridge to CTOR: This is a landmark study!

It establishes a mathematical bridge between neural network architectures and cellular automata. It demonstrates that introducing a global feedback parameter into standard cellular networks—which typically rely on local connections—enables the system to approximate deep

recurrent neural networks, thereby paving the way for the development of information-preserving algorithms.

7. Academic Conference Paper: "A Method for the Identification of Homogeneous Neural Networks via Activity Patterns"

- Original Citation: Adamatzky, A. I., Bronnikov, V. A. **On a method for identifying homogeneous neural networks based on a set of activity patterns** // In: Problems of Neurocybernetics (Proceedings of the IX All-Union Conference on Neurocybernetics), Rostov-on-Don, 1989, p. 344.
 - Abstract: Investigates the problem of mapping data-flow paths within uniform neural structures. The authors outline a non-invasive computational method to chart structural network links based solely on observed firing outputs and pattern states.
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Section 4: Hardware Innovation & Supercomputer Architecture

8. Technical Architecture Paper: "The LOCON Co-Processor: Modeling Systems with Local Interaction"

- Original Citation: Ivanov, S. F., Bronnikov, V. A., Solovyev, M. V., Adamatsky, A. I. **The LOCON Special-Purpose Processor: Modeling Systems with Local Interaction** // In: 1st All-Union Conference "Homogeneous Computing Environments and Systolic Structures," Lviv, 1990, Vol. 3, pp. 75–79.
- Abstract: This paper outlines the hardware architecture of the proprietary supercomputer project "LOCON" (developed at LKTB, Leningrad under the patronage of I.V. Berg). LOCON utilized a radically non-Von Neumann parallel computing matrix featuring up to 64,000 localized micro-processors designed explicitly for physical-level hardware acceleration of cellular automata computations and structural neural simulations (post Perceptron F. Rosenblatt).
- Bridge to CTOR: The algorithmic requirements to map multi-processor communication lanes without signal clashing on the LOCON grid led to the discovery of optimal topological structures—namely, boundaryless torus grids. The spatial game field of CTOR is a mathematical projection of the LOCON processor matrix layout.

9. Industrial Media Publications on Parallel Computing

- Original Citations: Bronnikov, V. A. **Supercomputer Production in the USSR** // PC Week RE (ITWeek), December 23, 1997.
 - Abstract: Historical and technical overviews documenting the development, hardware fabrication capabilities, and architectural paradigm of parallel computing infrastructure and specialized multi-processor matrices.
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Section 5: Mass Media, Public Validation & Modern Applications

10. Foundational Game Launch & Public Intellectual Outlets (1989)

- Original Citations: **Playing CTOR** // *Science and Life* (Nauka i Zhizn) popular science magazine, Gaming section, No. 9, 1989.
- **From the game "Life" to the game CTOR** // *PC Week RE* (ITWeek), 1989.
- Abstract: These landmark articles introduced CTOR (Cellular Torus) to the public mass media, demonstrating how John Conway's mathematical game *Life* could be elevated into a competitive, mathematically profound strategic framework running on boundaryless surfaces. The mechanics validated its capacity to train spatial-logical reasoning in human subjects.

11. Modern Implementations: Cognitive Training & Collective Intelligence (2015–2016)

- Original Citations / Multimedia Documentation:
 - Bronnikov, V. A. **Educational Course "Creativity and Memory"** // Discovery Academy, Richmond Hill, Ontario, Canada, 2015.
 - Bronnikov, V. A. **How to Measure Collective Intelligence** // TEDx Vaughan Lecture, Toronto, Canada, 2016.
- Abstract: Данная работа связывает исторические исследования в области математического моделирования — на стыке аппаратного и программного обеспечения — с прикладной когнитивистикой. Привлекая группы людей к выполнению задач с использованием заранее заданных цифровых и настольных матриц CTOR в условиях жестких временных ограничений, исследователи успешно продемонстрировали метод измерения когнитивного переключения между левым и правым полушариями (режимами логического мышления и распознавания образов), а также формализовали метрики коллективной интуиции и эффективности командного взаимодействия.
- Bridge to CTOR: A course based on the CTOR game, conducted at a school in Canada, explored team-versus-team gameplay with students aged 10–12. Using an online platform hosted on Azure, experiments were carried out to identify the most effective three-player teams. All decisions were made without communication between players, thereby modeling the mechanism of intuitive group decision-making.

Section 6: Comprehensive Monograph — Chapter-by-Chapter Historical Text & Modern Synthesis

12. Book Monograph: "Artificial Constraints of Natural Intelligence, or New Games for Children and Adults"

- Original Citation: Bronnikov, V. A. **Artificial Limitations of Natural Intelligence, or New Games for Children and Adults**. (Original Manuscript: 1990; Revised & Expanded Anniversary Edition: 2027).
- Abstract: This seminal monograph serves as the theoretical and philosophical manifesto underpinning the mathematical framework of the CTOR system. Written at the dawn of the personal computing era, the text bridges computational biology, cognitive neuroscience, and parallel computing architectures. It posits that strategic game environments, when freed from

artificial spatial boundaries, function not as entertainment, but as rigorous multi-agent laboratories capable of simulating the non-linear dynamics of complex natural ecosystems, evolutionary biology, and swarm intelligence.

Chapter 1: Preface

- 1990 Core Paradigm: Establishes the core dual-paradigm of the book: first, the rising socio-cognitive role of advanced intellectual games driven by the mass proliferation of personal computers; second, the rapid mathematization of natural sciences, using computer-driven models to decode complex biological frameworks.
- 2027 AI Convergence & CTOR Bridge: Directly maps to the current global focus on gamified environments as training grounds for AI. Just as DeepMind used Chess and Go to advance deep learning, this chapter establishes the game matrix as an essential instrument for mathematical modeling of living systems.

Chapter 2: Games Around Us

- Sub-Chapter 2.1: **The Game of Cognition**
 - 1990 Core Concept: Translating empirical natural sciences into Game Theory. Moving from physical experimentation to analytical modeling, and ultimately to computational simulation of ecosystems. Proposing that a process model formatted as a strategic game offers the most optimal algorithmic description of a parallel distributed process, laying the groundwork for advanced Cellular Automata (CA).
 - 2027 AI Convergence & CTOR Bridge: This anticipated the core architecture of Multi-Agent Reinforcement Learning (MARL). Modern AI research confirms that complex biological and computational systems are best modeled as parallel, game-theoretic environments.
- Sub-Chapter 2.2: **Intellectual Games and Society**
 - 1990 Core Concept: Examines the socio-cultural footprint of strategic gaming, exploring the crossover between operational industrial activity, abstract science, and art, concluding that the societal reliance on gamified cognitive processing is expanding.
 - 2027 AI Convergence & CTOR Bridge: Intersects with modern organizational behavioral models and decentralized coordination networks, validating the use of games to structure group interactions.
- Sub-Chapter 2.3: **Computer Games**
 - 1990 Core Concept: Outlines a proprietary classification of digital games, highlighting the most prospective vectors for educational deployment and cognitive research.

- 2027 AI Convergence & CTOR Bridge: Maps directly to computational pedagogy and adaptive training simulators used to augment human skill acquisition.

Chapter 3: **Psychological Aspects of Intellectual Games**

- Sub-Chapter 3.1: **Image-Based vs. Logical Processing in Cognition and Machines**

- 1990 Core Concept: Synthesizes psychology, neurophysiology, and cybernetics to contrast serial sequential processing with parallel/associative spatial thinking. Analyzing why specific game topographies are computationally trivial for machines while others present immense bottlenecks due to pattern recognition requirements.
- 2027 AI Convergence & CTOR Bridge: This directly targets the structural limitations of early symbolic AI. Today's deep learning revolution (specifically Convolutional Neural Networks and Vision Transformers) was built precisely to address the parallel, associative spatial processing constraints identified in this chapter.

- Sub-Chapter 3.2: **One Brain is Good, But Two Are Better**

- 1990 Core Concept: Maps cognitive mechanics onto brain bilaterality (Left vs. Right hemisphere functions). Formulating the fundamental hypothesis: *Intellectual games can be classified by their exact operational trigger of logical vs. image-based cognition.* Proposing that new games should feature adjustable rules capable of dynamically dialing or dosing the ratio between logical analysis and spatial-pattern intuition based on the user's cognitive state.
- 2027 AI Convergence & CTOR Bridge: Bridges directly to modern Neuro-Symbolic AI (combining neural networks for pattern recognition with symbolic logic for reasoning). CTOR serves as a live, physical benchmark that forces the player to constantly shift states between logical calculations and intuitive visual-spatial pattern assessment under tight time constraints.

Chapter 4: **New Games for Children and Adults**

Sub-Chapter 4.1: **How Games Are Invented**

1990 Core Concept: Details the explicit transition from a biological algorithm to an active intellectual game. Outlines the precise rules of CTOR, demonstrating how the interactions of opponents replicate the fundamental cosmic interplay of Order and Chaos, leading to finite, goal-oriented system endings.

- 2027 AI Convergence & CTOR Bridge: Represents the translation of self-organizing mathematical models into gamified environments, allowing researchers to study systemic entropy and stability in real time.

Sub-Chapter 4.2: **What Living Systems Play**

- 1990 Core Concept: Demonstrating that individual agents in both biological networks and strategic games operate under identical mathematical transition laws. Highlighting the low parallel processing capacity of classical games (e.g., Chess) versus the high-parallelism

of homogeneous automaton networks. Formulating a radical conclusion: *The neural algorithms used to solve an intellectual problem and the algorithms driving an ecological simulation are isomorphic descriptions of the same fundamental process.*

- 2027 AI Convergence & CTOR Bridge: Directly maps to Swarm Intelligence and Artificial Life (Alife) research. CTOR proves that abstract game strategies match the mathematical modeling of emergent behavior seen in biological populations or cellular tissue development.

Sub-Chapter 4.3: **Limitations of Machine Intelligence**

- 1990 Core Concept: Assesses computer capabilities, stating that success in executing biological simulations depends entirely on the architectural alignment between localized model operations and the computer's memory structure. Breaks down the first three computer algorithms developed to run CTOR, drawing a direct analogy between CTOR logic and parallel algorithms for biological modeling, introducing the crucial concept of micro- and macro-variables.
- 2027 AI Convergence & CTOR Bridge: Deeply relevant to the Von Neumann Architecture Bottleneck and the move toward modern Neuromorphic Chips. It anchors CTOR as a benchmark specifically built to test parallel, decentralized data routing—mimicking how modern graph neural networks handle localized vector spaces.

Chapter 5: **Where the Game Ends and Science Begins**

- Sub-Chapter 5.1: **Natural Constraints of Artificial Intelligence**

- 1990 Core Concept: Critiquing classical AI optimization. Stating that computer architectures suffer from an inherent "anti-parallelism" and are fundamentally misaligned with the natural biological processes they attempt to model. Arguing against the hyper-optimistic claims of early "expert system" developers, asserting that true artificial intelligence must evolve out of natural, structural, and structural-parallel machine naturalness rather than raw brute-force computing.
- 2027 AI Convergence & CTOR Bridge: This chapter perfectly aligns with the contemporary AI Safety, Ethical Alignment, and Artificial General Intelligence (AGI) discourse. By questioning the "naturalness" of silicon-based sequential logic compared to evolutionary biology, it frames CTOR not just as a game, but as an evaluative, organic testing matrix.

- Sub-Chapter 5.2: **Artificial Constraints of Natural Intelligence**

- 1990 Core Concept: Analyzing human cognitive boundaries. While the brain is optimized for local survival, its abstract thinking engine is prone to constraints when dealing with the hyper-complexity of non-linear biological networks. Nature does not prioritize optimization in a narrow mathematical sense. Even on paths of high abstraction, definitive biological results remain elusive because biology lacks the neat theoretical symmetry of physics. This asks a critical epistemological question: *Does our reliance on rigid, axiomatic, linear logic restrict our understanding of nature? Can we bypass these*

constraints by shifting from axiomatic theories to Constructive Mathematics to decode fundamental evolutionary processes?

- 2027 AI Convergence & CTOR Bridge: This positions your work alongside modern critics of pure deep-learning scale. It argues that raw scale cannot capture emergent biological truths without embedded structural mathematics—a foundational pillar of modern Neuro-Symbolic AI and Constructive Mathematical Biology.
- Sub-Chapter 5.3: **Toward Computational Biology**
 - 1990 Core Concept: Formally defining the frontier of "Computational Biology." Outlining a methodology to synthesize macroscopic system dynamics from simple microscopic local transition rules (bridging micro- and macro-theories). This connects the complexity of living behaviors with simplicity of organization. It answers how—knowing the properties of individual elements (individual organism, cell in an embryo, neuron, mediator molecule) and environmental macro-states—one can dynamically manipulate and control macro-states of an ongoing computational experiment through micro-element parameters.
 - 2027 AI Convergence & CTOR Bridge: Directly maps to modern generative life simulations, multi-agent frameworks, and computational biology labs studying emergent cellular and neural behaviors.
- Sub-Chapter 5.4: **Global Games**
 - 1990 Core Concept: Formulating a prophetic distributed computing architecture: *Designing an engaging, scalable strategic game engine that simultaneously models localized segments of a massive biological ecosystem.* Distributing this framework globally across millions of personal computers with varied local transition parameters, and aggregating user-generated telemetry data from discs or networks into a central parallel supercomputer to redefine macro-steps of evolutionary modeling.
 - 2027 AI Convergence & CTOR Bridge: This is a literal 1990 prediction of Crowdsourced Distributed Citizen Computing (e.g., Foldit) combined with modern Decentralized Edge AI Training. It showcases CTOR's ultimate scalability—deploying mass-scale game networks involving thousands of simultaneous actors to generate massive, empirical telemetry datasets for collective intelligence and complex system behavioral modeling.